

Data Converters

Exercise 04 – Settling Analysis of a Charge Redistribution Track & Hold Circuit

Shown in Figure 1 is a charge redistribution track & hold circuit whose settling and noise behavior is to be analyzed. The equivalent circuit at the beginning of the sampling phase Φ_1 and redistribution phase Φ_2 is shown in Figure 2.

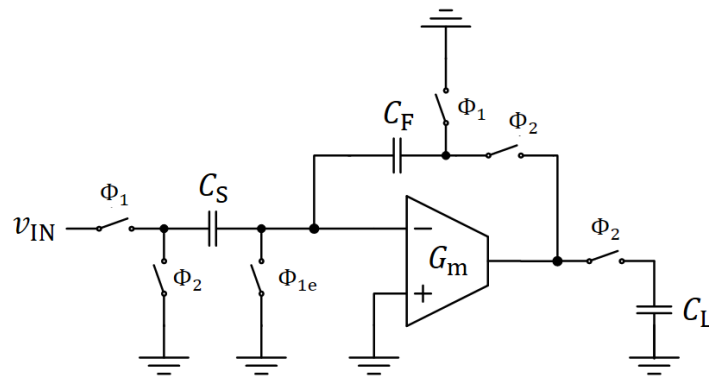


Figure 1: Charge redistribution track & hold circuit.

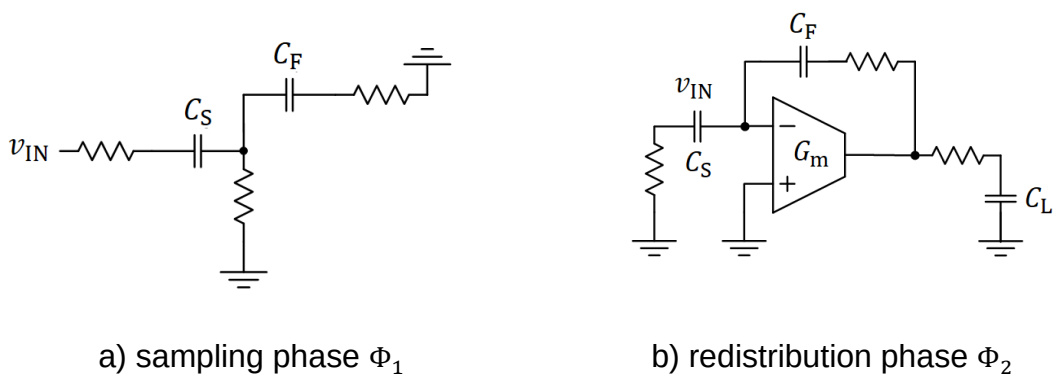


Figure 2: Equivalent circuit at the beginning of the sampling phase Φ_1 (a) and redistribution phase Φ_2 (b).

The settling behavior of the sampling phase Φ_1 was discussed in the lecture and is determined by the sampling capacitor C_S and the on-resistance of the switch driven by Φ_{1e} .

Considering the redistribution phase Φ_2 , the amplifier and the capacitors determine the settling behavior as will be elaborated in the discussion of the exercise. All resistors can thus be shorted for the analysis of the settling behavior. Moreover, instead of redistributing the charge stored on the sampling capacitor C_S due to its pre-charging to v_{IN} in the sampling phase Φ_1 (see Figure 2.b), it is equivalent to drive the discharged sampling capacitor C_S by a voltage source generating a step with amplitude v_{IN} in the redistribution phase Φ_2 . The equivalent circuit is shown in Figure 3.

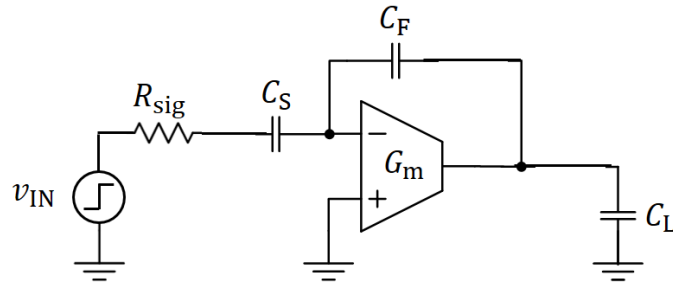


Figure 3: Equivalent circuit for analyzing the settling behavior in the redistribution phase.

The small-signal equivalent circuit of the G_m -cell to be used in the analysis is given in Figure 4.

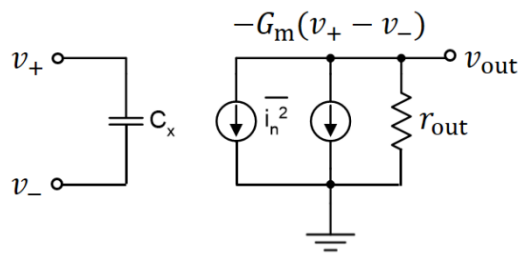


Figure 4: Small-signal equivalent circuit of the G_m -cell.

Task I – Determination Transfer Function – Nodal Analysis

- Draw the small-signal equivalent circuit of Figure 3 using the small-signal equivalent circuit of the G_m -cell shown in Figure 4. In the process, neglect the current noise source $\overline{i_n^2}$ of the G_m -cell and replace the input voltage source v_{in} together with its resistance R_{sig} by its equivalent Norton current source.
- Perform a nodal analysis and derive the transfer function $TF(s) = v_{out}/v_{in}$ by means of a technical computing software, e.g., Mathematica or Maple.
- Simplify your result for $g_{out} = 1/r_{out} \ll G_m$.
- How large is the DC gain of the transfer function $TF(s)$, i.e., $TF(s = 0)$?
- Determine the zero of the transfer function $TF(s)$.
- Considering the small-signal model of the circuit, what is your intuitive guess of the pole at v_{out} ?
- Determine the pole of the transfer function $TF(s)$. Does the result match your intuitive result? How can the mismatch be explained?
- State the time-domain step response $TF(t)$ neglecting the zero.

Task II – Determination Transfer Function – Asymptotic Gain Formula

While a nodal analysis may be applied to any arbitrary linear circuit and thus to any small-signal equivalent circuit in order to determine the voltages of its nodes, it hardly helps in building up intuitive understanding of a circuit's electrical behavior. Particularly, if a feedback system is analyzed as in the considered example. For such systems, the derivation of the transfer function according to the asymptotic gain formula is much more insightful.

- Watch the video “Analog Electronics 2 – Loop Gain Analysis – Asymptotic Gain Formula” by David Johns.
- Draw the small-signal equivalent circuit of Figure 3 using the small-signal equivalent circuit of the G_m -cell shown in Figure 4. In the process, neglect the current noise source $\overline{i_n^2}$ of the G_m -cell and replace the resistor R_{sig} by a short.
- Determine the loop gain $L(s)$ by means of a technical computing software, e.g., Mathematica or Maple.
- Simplify the result of the loop gain $L(s)$ for $g_{out} = \frac{1}{r_{out}} \gg \text{any capacitance } C_i$.

- e. Compare the simplified loop gain $L(s)$ to $\beta * G_m * r_{out}$ and show that the return factor β results in $C_F / (C_F + C_S + C_x)$. Which information does β provide?
- f. Determine the infinite loop gain transfer function $A_{L\infty}(s)$ by means of a technical computing software, e.g., Mathematica or Maple.
- g. Determine the zero loop gain transfer function $A_{L0}(s)$ by means of a technical computing software, e.g., Mathematica or Maple.
- h. Determine the closed-loop transfer function $TF_{AGF}(s)$ by means of a technical computing software, e.g., Mathematica or Maple, and compare it to $TF(s)$ derived in task I.
- i. Show that the output impedance of the feedback system according to Figure 5 results in

$$r_o = \left. \frac{v_{out}}{i_{out}} \right|_{v_{in} = 0 \text{ V}} \approx \frac{1}{\beta G_m} \parallel \frac{1}{s(1 - \beta)C_F}$$

where β equals the return factor derived in task II.e.

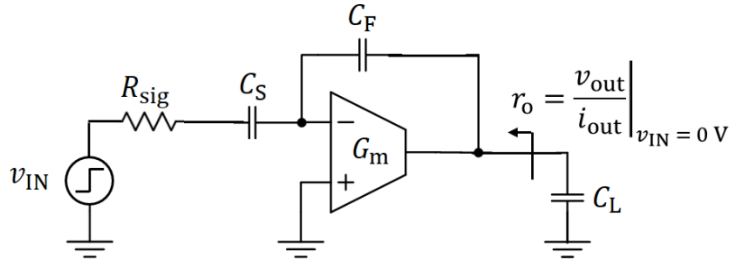


Figure 5: Definition of the output resistance r_o of the feedback system.

- j. Show that the pole at the output node results in

$$p = \frac{\beta G_m}{C_{Ltot}} = \frac{\beta G_m}{C_L + (1 - \beta)C_F}$$

Task III – Dynamic and Static Settling Error (without Slewing)

According to the asymptotic gain formula, the closed-loop transfer function of a feedback system results in

$$TF(s) = \frac{out(s)}{in(s)} = \frac{A_{L\infty}(s)L(s)}{1 + L(s)} + \frac{A_{L0}(s)}{1 + L(s)}$$

Neglecting the bypass signal path $A_{L0}(s)$, which is anyway a high-pass filter in the considered circuit, the low to mid-frequency transfer function results in

$$TF(s) = \frac{out(s)}{in(s)} \approx \frac{A_{L\infty}(s)L(s)}{1 + L(s)}$$

- a. Determine the static settling error ε_s defined as

$$\varepsilon_s = \frac{TF(s) - TF_{\text{ideal}}(s)}{TF_{\text{ideal}}(s)}$$

where

$$TF_{\text{ideal}}(s) = TF(s)|_{L(s) \rightarrow \infty} = A_{L\infty}(s)$$

- b. Determine the dynamic settling error ε_d defined as

$$\varepsilon_d = \frac{TF(t) - TF_{\text{ideal}}(t)}{TF_{\text{ideal}}(t)}$$

where

$$TF(t) = \text{your result task 1.h}$$

$$TF_{\text{ideal}}(t) = TF(t)|_{t \rightarrow \infty}$$

Task IV – Dynamic and Static Settling Error (with Slewing)

It's your lucky day! This task will be presented and discussed in the exercise.